ECE 340: Semiconductor Devices

Lecture 13: Generation

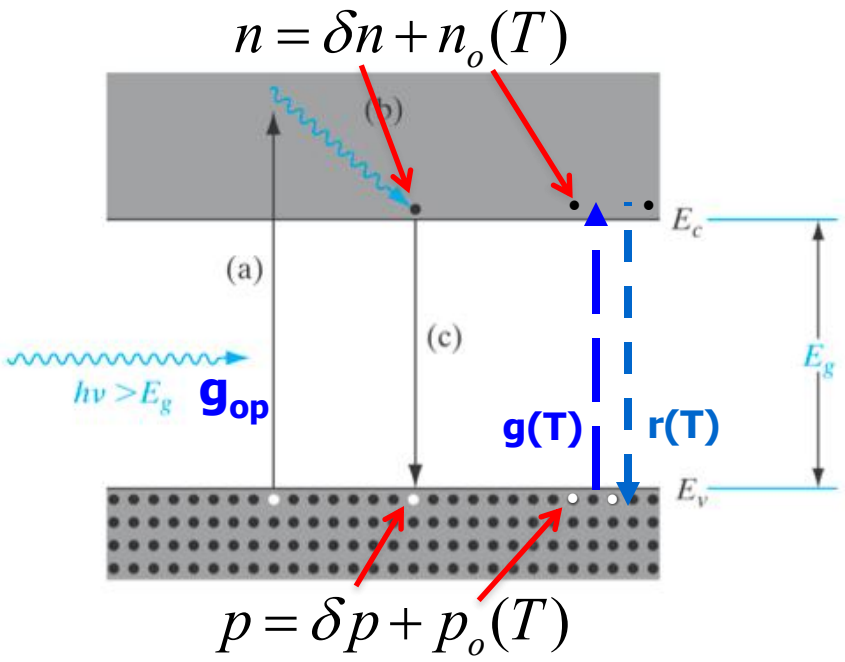
Prof. Hyunseok Kim

Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

E-mail: hyunseok@illinois.edu

Office Hours: Thursdays 12p-1p, 2256 HMNTL

(Recap) Optical Absorption of Photons and Excess Carriers

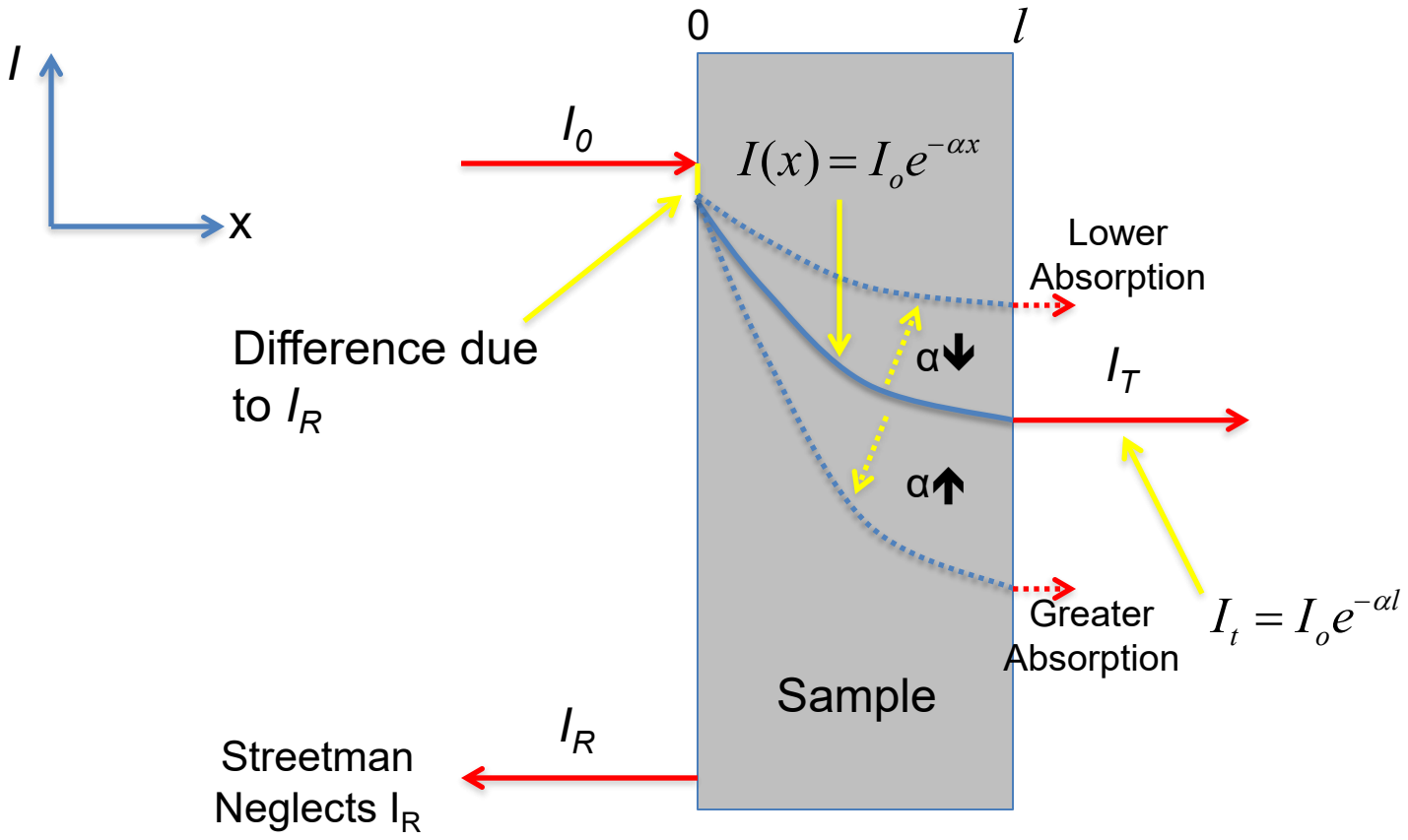


- A photon with energy $hv<E_g$ will not be absorbed. Hence, it is transmitted through the sample.
  - Insulators are frequently transparent due to their large bandgap
- A photon with energy $hv>E_g$ can be absorbed to excite an electron from the valance band to the conduction band.
- Thermal generation of carriers provides the equilibrium carrier concentrations, $n_o(T)$ and $p_o(T)$
- Optical generation of excess carriers (δn and δp) can contribute to the conductivity

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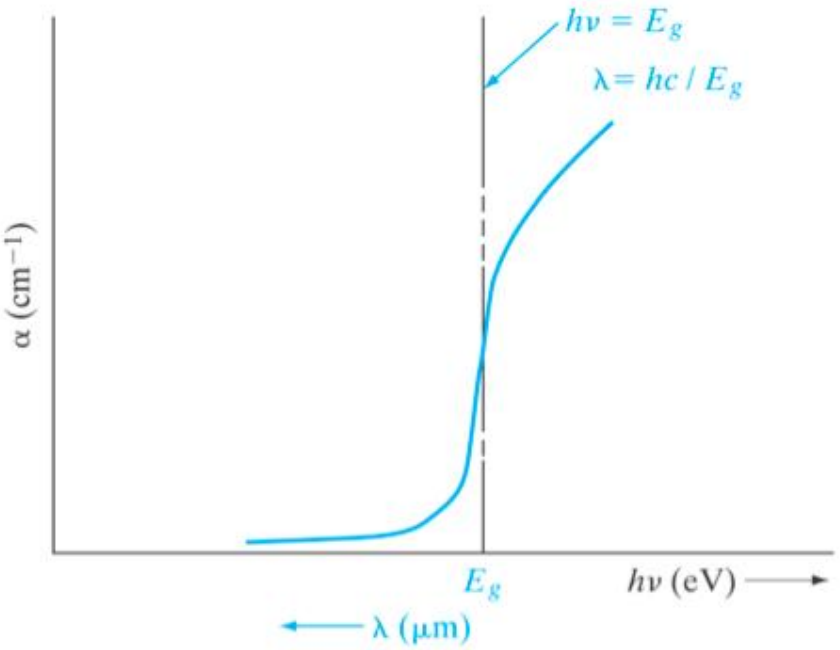
(Recap) Photon Flux Versus Position

- The photon flux drops exponentially in the sample
- The relationship for the generated carriers is more complicated due to diffusion



(Recap) Absorption Coefficient

- The absorption coefficient depends upon the incident wavelength λ
- Below the bandgap there is minimal absorption
- The units of absorption are typically cm^{-1}



Useful Relationship:

$$E(eV) = h\nu = h\frac{c}{\lambda} = \frac{1.239}{\lambda(\mu m)}$$

A photon with a wavelength of 1 μm has an energy of 1.239 eV

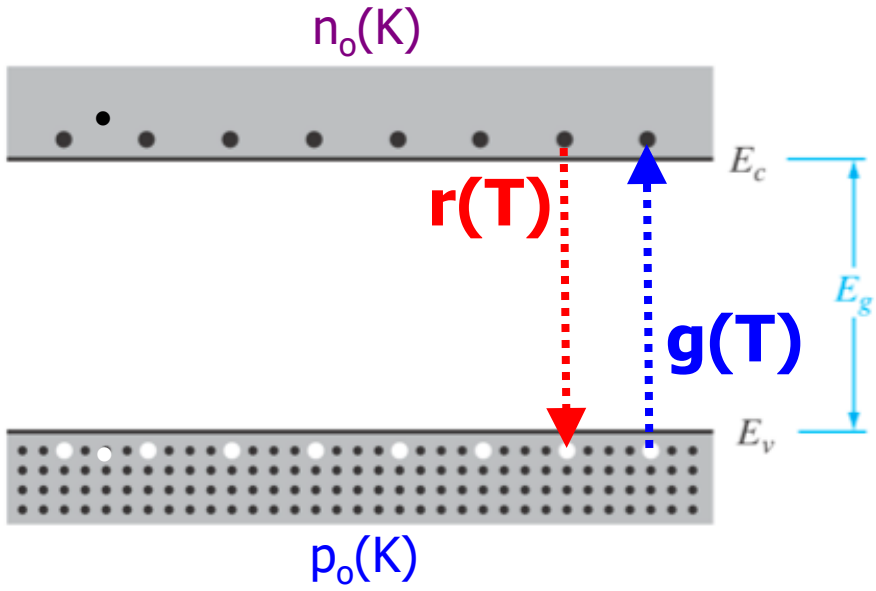
Generation

Terminology Reminder

- ***Equilibrium:*** A special case of steady state. No external perturbing excitation is present except temperature, and there is no net motion of charge.
- ***Steady State***: A condition in which all perturbing processes are constant. In steady state, opposing processes are present that balance perturbing processes.

Equilibrium Generation

Some electrons have sufficient energy to jump to the conduction band (phonon interactions)



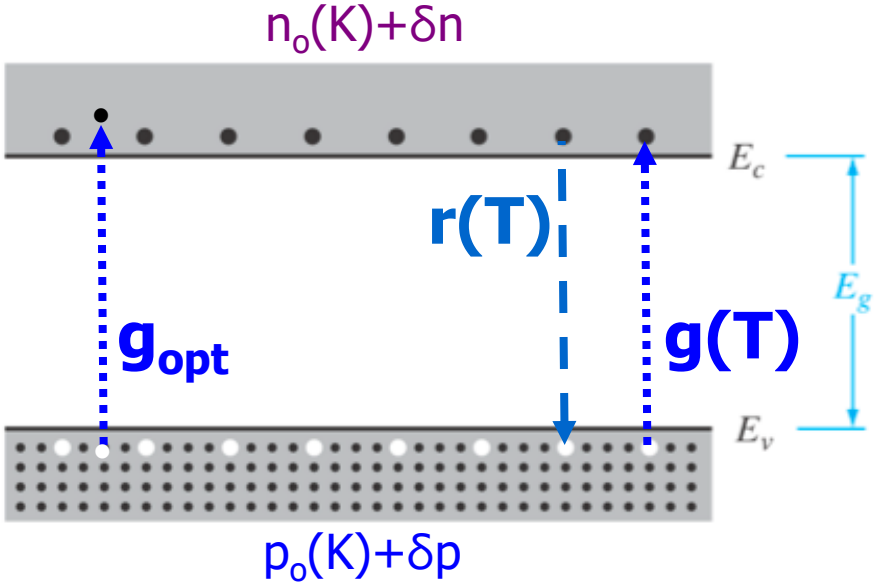
Generation and recombination are balanced:

$$g(T) = g_i = r_i = \alpha_r n_o p_o = \alpha_r n_i^2$$

α_r : e-h recombination coefficient
(Not the same as the absorption coefficient α)

Steady Illumination

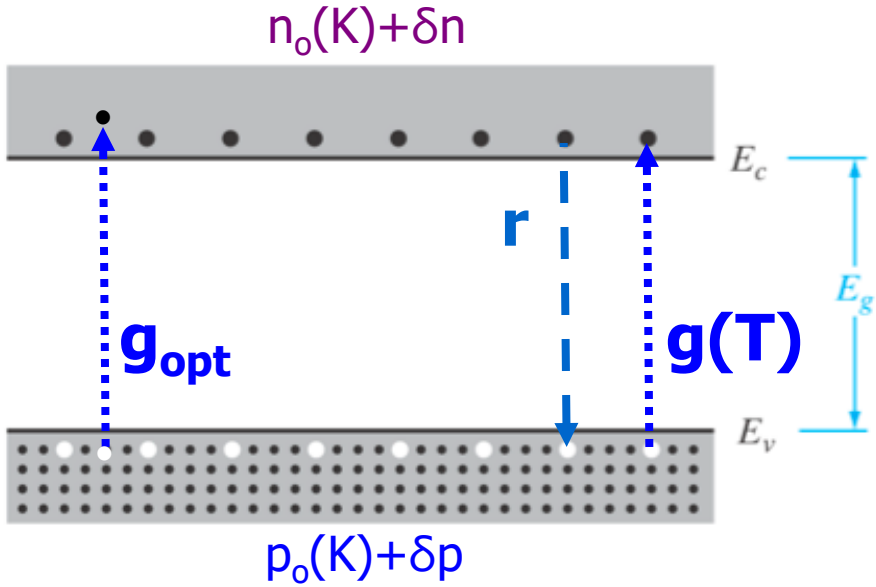
- Under steady illumination, an optical generation rate g_{opt} is added to the equilibrium generation rate *g(T)* and is balanced by the recombination rate *r(T)*
  - Or: $r=g(T)+g_{opt}$
- An additional population of electrons and holes $\delta n=\delta p$ will be produced by illumination
- If the illumination stops, $g_{opt}=0$ but recombination occurs at a faster rate than generation until the excess carriers are depleted



$$r = \alpha_r np = \alpha_r (n_o + \delta n)(p_o + \delta p)$$
$$= \alpha_r [n_o p_o + n_o \delta p + p_o \delta n + \delta n \delta p]$$
$$= \alpha_r [n_o p_o + (n_o + p_o)\delta n + \delta n^2]$$

Steady State Generation

$$g = g(T) + g_{opt} = r = \alpha_r np$$
$$= \alpha_r (n_0 + \delta n)(p_0 + \delta p)$$
$$= \alpha_r \left[n_0 p_0 + n_0 \delta p + p_0 \delta n + \delta n \delta p \right]$$
$$= \alpha_r \left[n_0 p_0 + (n_0 + p_0)\delta n + \delta n^2 \right]$$
$$= \alpha_r n_0 p_0 + \alpha_r \left[(n_0 + p_0)\delta n + \delta n^2 \right]$$



Recall: $g(T) = \alpha_r n_i^2 = \alpha_r n_0 p_0$

so

$$g_{op} = \alpha_r \left[(n_0 + p_0)\delta n + \delta n^2 \right] \approx \alpha_r (n_0 + p_0)\delta n = \frac{\delta n}{\tau_n}$$

$$\tau_n = \frac{1}{\alpha_r (n_0 + p_0)}$$

and

$$\delta n = \delta p = g_{op} \tau_n$$

Assume *δn* is small compared to majority carrier concentration

Important Definition

- ***Minority Carrier Lifetime*** or ***Recombination Lifetime***: the time constant by which the excess carrier population decays to the equilibrium value

$$\tau_n = \frac{1}{\alpha_r \left(n_0 + p_0 \right)}$$

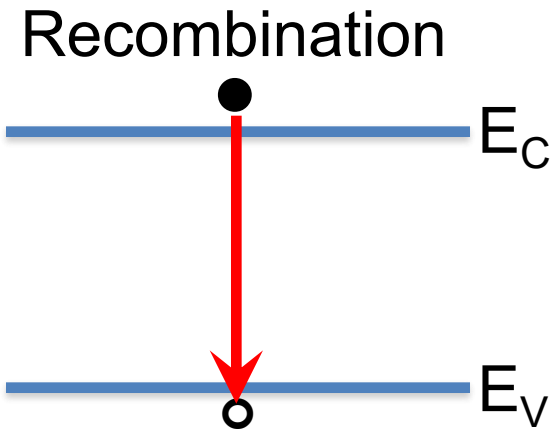
Recombination

Definitions

- ***Photoluminescence:*** Light emission that occurs after optical excitation of carriers
- ***Cathodoluminescence:*** Light emission that occurs after carriers have been excited by high-energy electron bombardment
- ***Electroluminescence:*** Light emission that occurs after carriers have been introduced using an electrical current

Band-to-Band Recombination

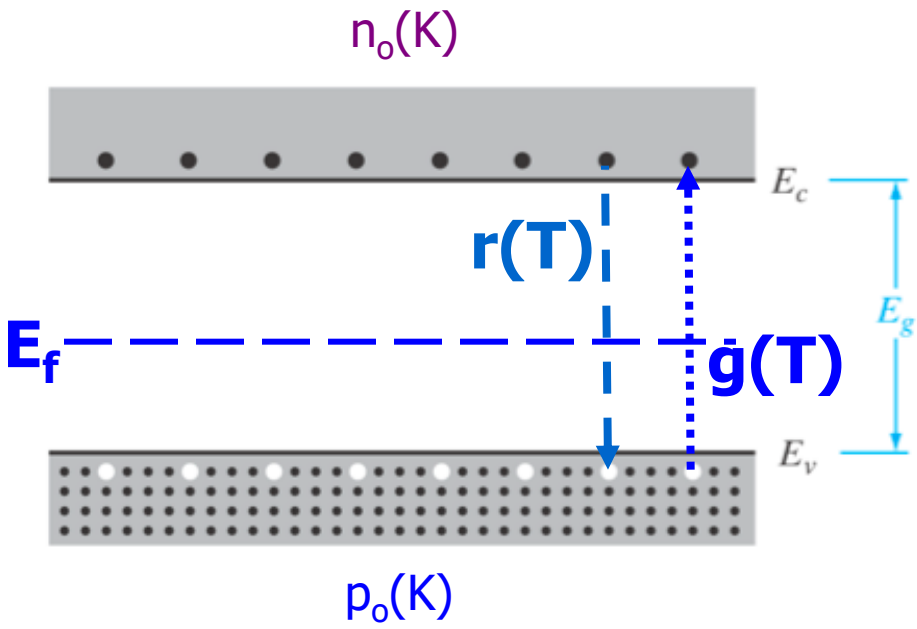
- For band-to-band electron-hole recombination, the "reaction rate" is proportional to the product of the concentration of electrons and holes:
  - $R_n=R_p=\alpha_r np$
  - α_r is the ***bimolecular recombination coefficient***, also called the capture coefficient
  - R_n and R_p are the number of electrons and holes recombining, respectively, per unit volume per unit time.



$$R_n = R_p = \alpha_r np$$

Reminder: Equilibrium Condition

The rate of recombination is proportional to the number of electrons in the conduction band times the number of holes in the valence band



$$r(T) = g(T) = \alpha_r n_i^2$$

- For any semiconductor (n-type, p-type, intrinsic) in equilibrium, the thermal generation of electron-hole pairs (EHP) is balanced by the recombination rate
  - Or: r(T)=g(T)
- Hence, the equilibrium electron and hole concentrations, $n_o(T)$ and $p_o(T)$ are maintained

$$r(T) = \alpha_r n_o(T) p_o(T)$$

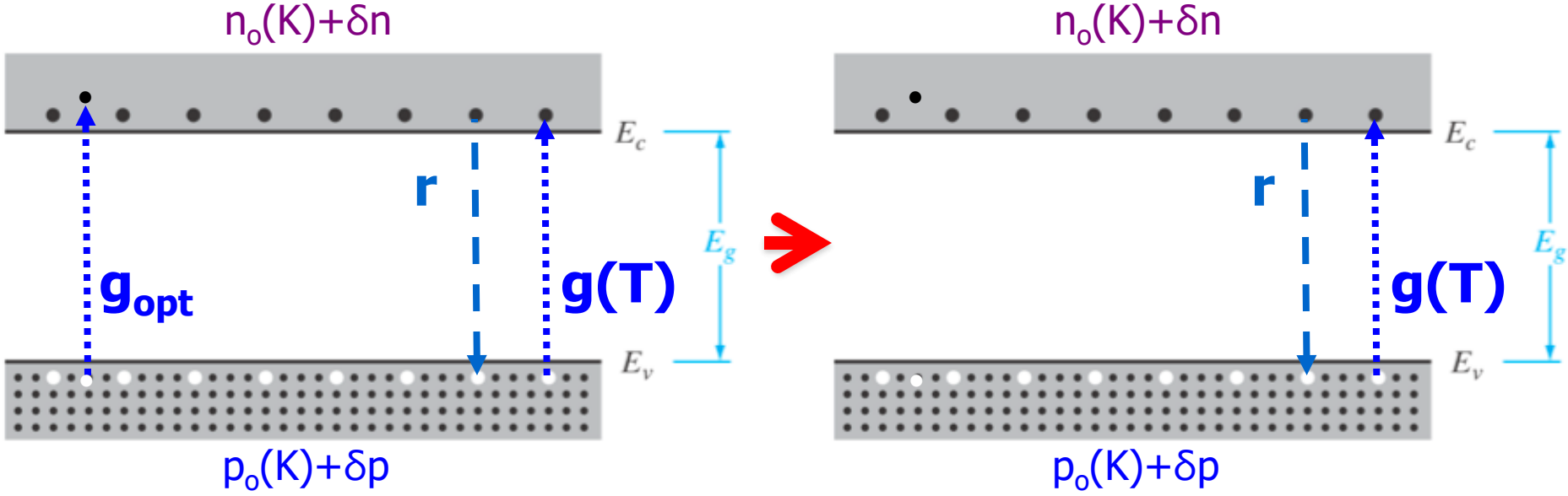
$$g(T) = \alpha_r n_i(T) n_i(T)$$

$$n_o(T) p_o(T) = n_i^2(T)$$

- The Fermi Level is used to quantify the electron and hole concentrations

What Happens When the Steady Illumination Stops?

Optical Generation Stops: $g_{opt}(t=0)=0$



Recombination: $r = g(T) + g_{opt}$

Generation: $g(T) + g_{opt}$

Generation and recombination are balanced

Recombination: $r = g(T) + g_{opt}$

Generation: $g(T)$

Generation and recombination are not balanced

The number of excess carriers decreases:

$$\frac{dn(t)}{dt} = g(T) - r(t) = \alpha_r n_i^2 - \alpha_r n(t) p(t)$$

A Little Bit of Math:

g(T) r

$$\frac{dn(t)}{dt} = \alpha_r n_i^2 - \alpha_r n(t)p(t)$$

$$= \alpha_r n_i^2 - \alpha_r \left[n_0 + \delta n(t)\right]\left[p_0 + \delta p(t)\right]$$

$$= \alpha_r n_i^2 - \alpha_r \left[n_0 + \delta n(t)\right]\left[p_0 + \delta n(t)\right]$$

$$= \alpha_r n_i^2 - \alpha_r \left[n_0 p_0 + \left(n_0 + p_0\right)\delta n(t) + \delta n^2(t)\right]$$

$$= -\alpha_r \left[\left(n_0 + p_0\right)\delta n(t) + \delta n^2(t)\right]$$

We can now look at specific cases: n-type, p-type, etc.

Solving the Differential Equation

p-type Material

1) Assume low-level injection:

neglect $\delta n^2(t)$

2) p-type material so $p_0 >> n_0$

$$\frac{d\delta n(t)}{dt} = -\alpha_r\left[\left(n_0 + p_0\right)\delta n(t) + \delta n^2(t)\right]$$

$$\approx -\alpha_r\left(n_0 + p_0\right)\delta n(t) \approx -\alpha_r p_0 \delta n(t)$$

The solution is:

$$\delta n(t) = \Delta n e^{-\alpha_r p_0 t} = \Delta n e^{-t/\tau_n}$$

where

$$\Delta n = \delta n(t = 0)$$

$$\tau_n = \frac{1}{\alpha_r p_0} \quad \text{or}$$

$$\tau_n = \frac{1}{\alpha_r\left(n_0 + p_0\right)}$$

n-type Material

1) Assume low-level injection:

neglect $\delta p^2(t)$

2) n-type material so $n_0 >> p_0$

$$\frac{d\delta p(t)}{dt} = -\alpha_r\left[\left(n_0 + p_0\right)\delta p(t) + \delta p^2(t)\right]$$

$$\approx -\alpha_r\left(n_0 + p_0\right)\delta p(t) \approx -\alpha_r n_0 \delta p(t)$$

The solution is:

$$\delta p(t) = \Delta p e^{-\alpha_r n_0 t} = \Delta p e^{-t/\tau_p}$$

where

$$\Delta p = \delta p(t = 0)$$

$$\tau_p = \frac{1}{\alpha_r n_0} \quad \text{or}$$

$$\tau_p = \frac{1}{\alpha_r\left(n_0 + p_0\right)}$$

Important Definition (Again)

- ***Minority Carrier Lifetime*** or ***Recombination Lifetime***: the time constant by which the excess carrier population decays to the equilibrium value

$$\tau_n = \frac{1}{\alpha_r \left(n_0 + p_0 \right)}$$

Example: Recombination in p-GaAs

- Note: The assumptions $p_o >> n_o$ and $p_o >> \delta n$ are valid
- $dn=dp=10^{14}\ cm^{-3}$ has been shut off at t=0
- τ_n=10 ns

$$A \text{ GaAs sample } N_a = 10^{15}\,cm^{-3} = p_o$$

$$n_o = \frac{n_i^2}{p_o} = \frac{(10^6\,cm^{-3})^2}{10^{15}\,cm^{-3}} = 10^{-3}\,cm^{-3}$$

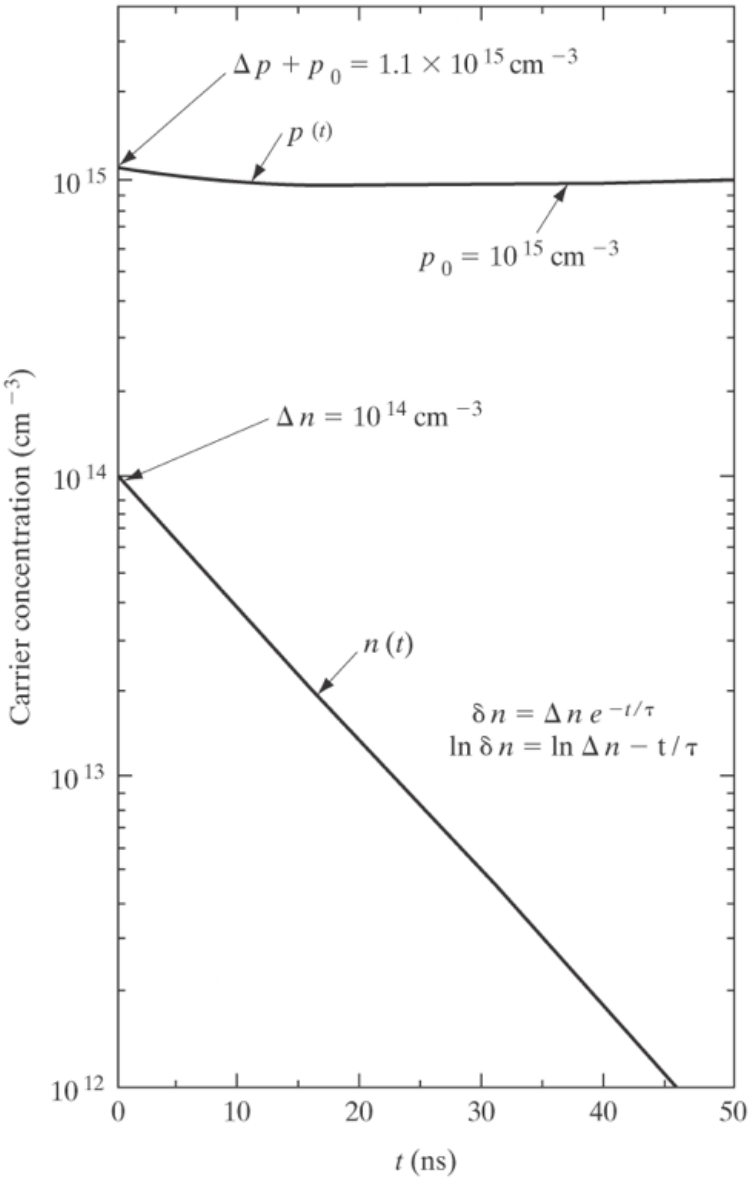
$$\delta n(t) = \Delta n e^{-t/\tau_n} = 10^{14}\,e^{-t/10^{-8}}\,cm^{-3}$$

$$n(t) = n_o + \delta n(t) = \delta n(t)$$

$$\delta p(t) = \delta n(t) = 10^{14}\,e^{-t/10^{-8}}\,cm^{-3}$$

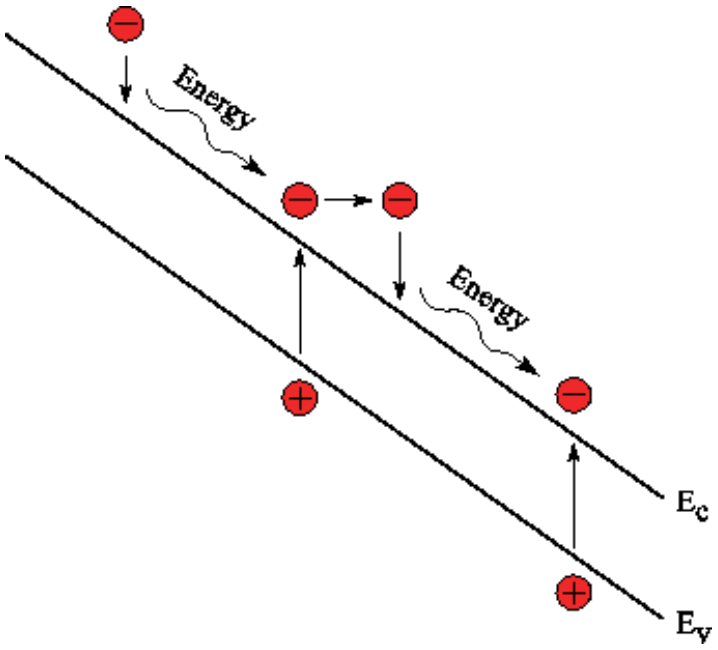
$$p(t) = p_o + \delta p(t)$$

$$p(t) = 10^{15}\,cm^{-3} + 10^{14}\,e^{-t/10^{-8}}\,cm^{-3}$$



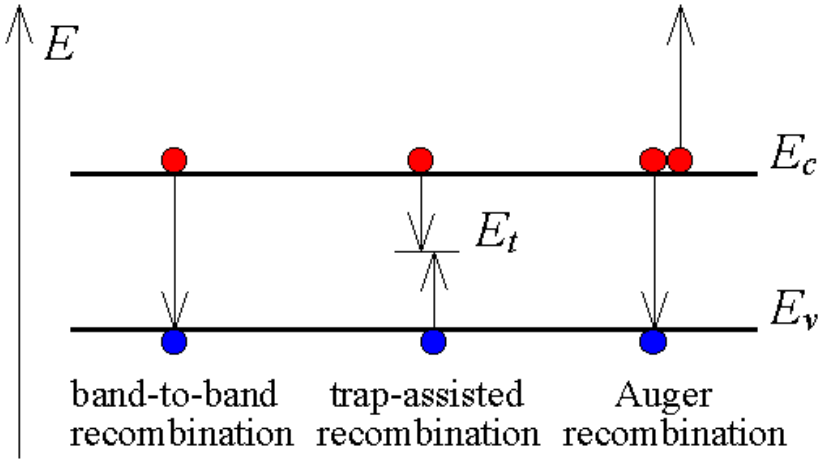
Other G-R Processes

Impact Ionization



http://www.iue.tuwien.ac.at/phd/entner/node11.html

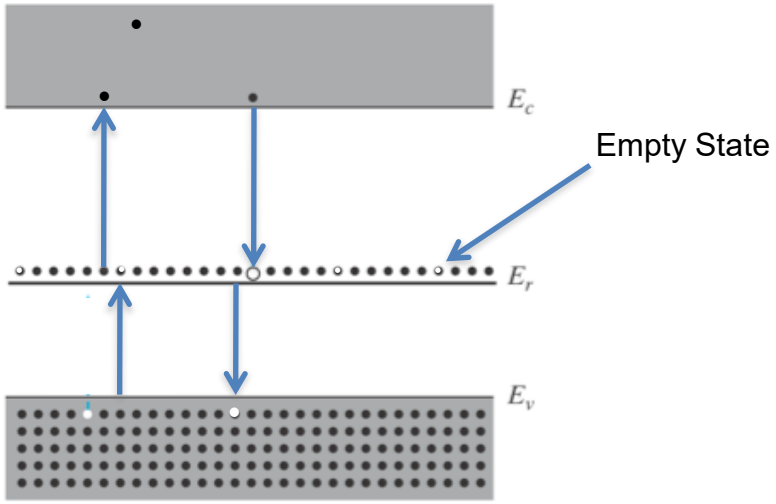
Auger, SRH GR



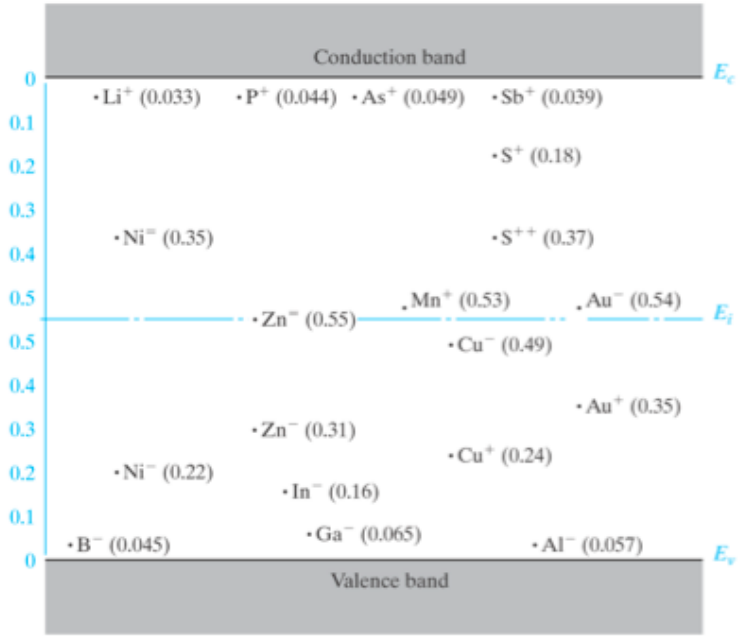
http://ecee.colorado.edu/~bart/ecen5355/newbook/chapter2/recomb.htm

G-R Through Trapping Centers (SRH)

Generation-Recombination



Traps in Silicon



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Quasi Fermi Levels

Steady Illumination: Revisited

$$r = \alpha_r np = \alpha_r (n_o + \delta n)(p_o + \delta p)$$
$$= \alpha_r [n_o p_o + n_o \delta p + p_o \delta n + \delta n \delta p]$$
$$= \alpha_r [n_o p_o + (n_o + p_o)\delta n + \delta n^2]$$
$$= g(T) + g_{opt}$$

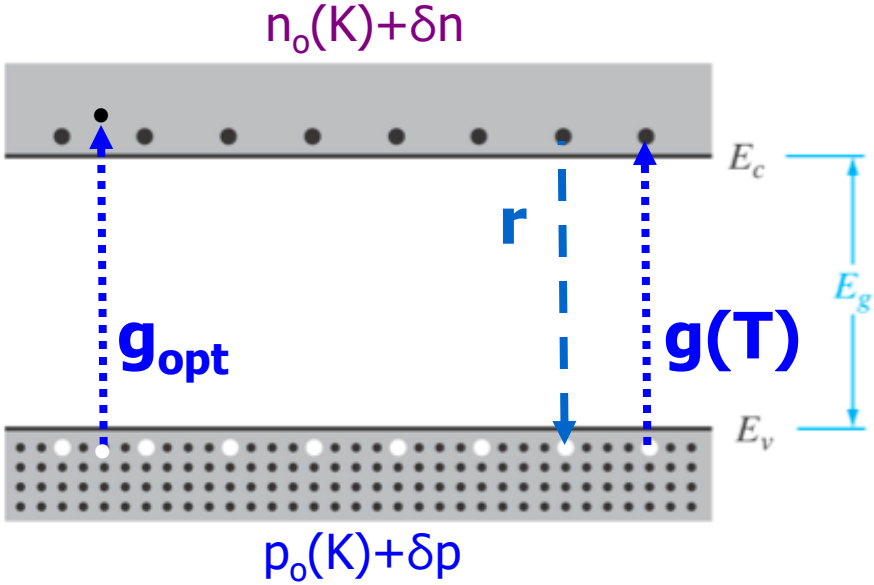
Recall: $g(T) = \alpha_r n_i^2 = \alpha_r n_0 p_0$

so

$$g_{op} = \alpha_r (n_0 + p_0)\delta n = \frac{\delta n}{\tau_n}$$

and

$$\delta n = \delta p = g_{op} \tau_n$$



What Happens to the Fermi Level?

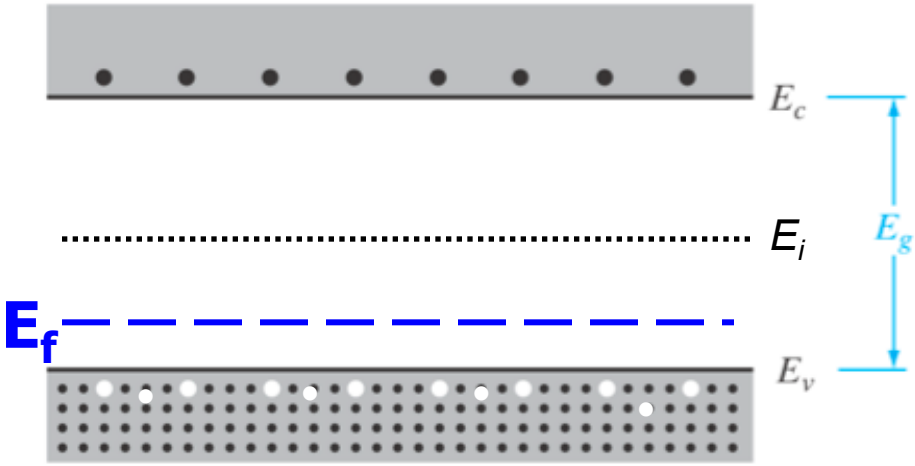
Quasi-Fermi Level

- A variety of mechanisms can create excess carriers (electrons and holes)
- The carriers may or may not be created in equal numbers
  - This depends upon the mechanism that created them
- Regardless of how they were created, the quantity of excess carriers can be described by integrating the density of states times the probability of occupancy
- The probability of occupancy is still a Fermi distribution, but with a modified Fermi energy, the “Quasi-Fermi Energy” or “Quasi-Fermi Level”

What Happens to the Fermi Level?

Equilibrium

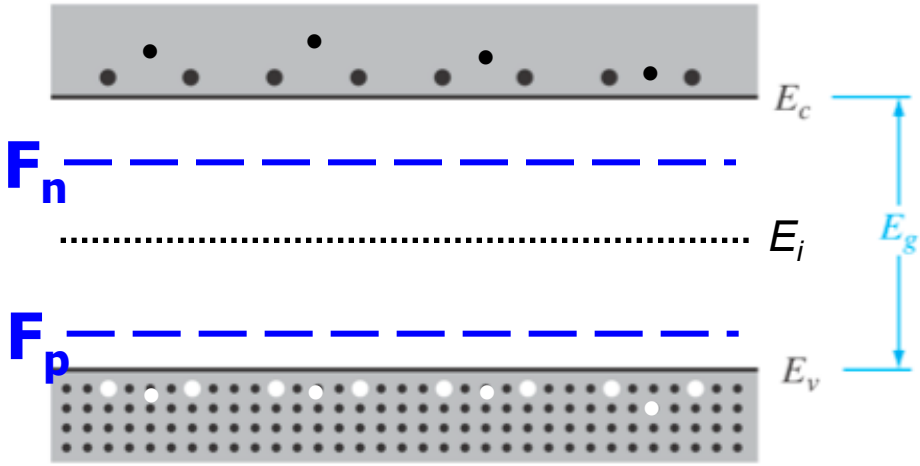
$n_o(K)$



$p_o(K)$

With Excess Carriers

$n_o(K)+\delta n$



$p_o(K)+\delta p$

The Fermi level in the presence of excess carriers is called the "quasi-Fermi level"

$$n_0 = n_i e^{(E_F - E_i)/kT}$$

$$p_0 = n_i e^{(E_i - E_F)/kT}$$

$$n = n_i e^{(F_n - E_i)/kT}$$

$$p = n_i e^{(E_i - F_p)/kT}$$

Quasi Fermi Levels

- Electron Quasi-Fermi Level:

$$n = n_i e^{(F_n - E_i)/kT} = n_i e^{(F_n - E_i + E_f - E_f)/kT}$$

$$= n_i e^{(E_f - E_i)/kT} e^{(F_n - E_f)/kT} = n_o e^{(F_n - E_f)/kT}$$

$$\delta n = n - n_o = n_o (e^{(F_n - E_f)/kT} - 1) = g_{op} \tau_n$$

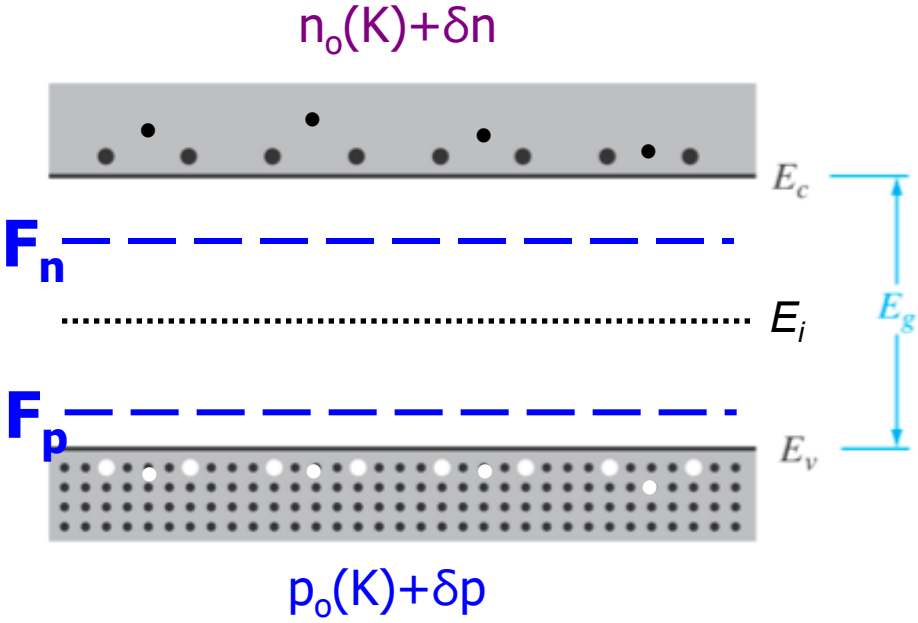
- Hole Quasi-Fermi Level:

$$p = p_o e^{(E_f - F_p)/kT}$$

$$\delta p = p - p_o = p_o (e^{(E_f - F_p)/kT} - 1) = g_{op} \tau_p$$

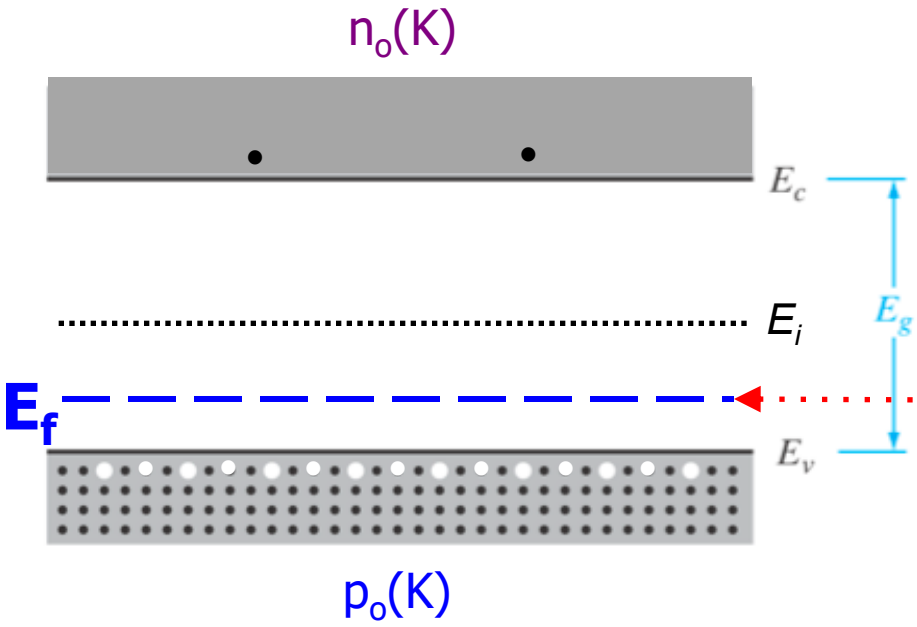
$$\delta n = \delta p = g_{op} \tau_n$$

Band Diagram, F_n and F_p

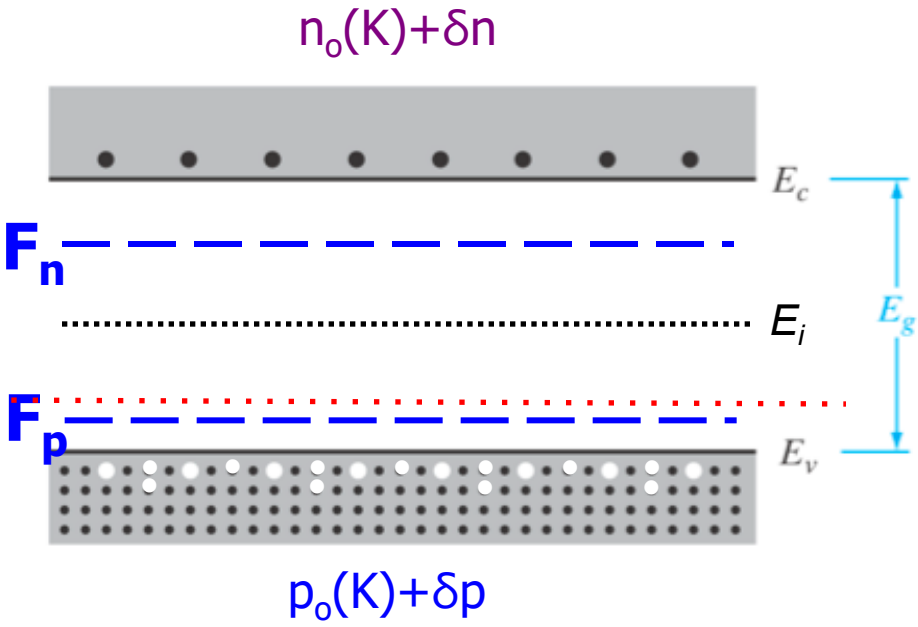


Quasi-Fermi Levels in p-type Material

Equilibrium



With Excess Carriers

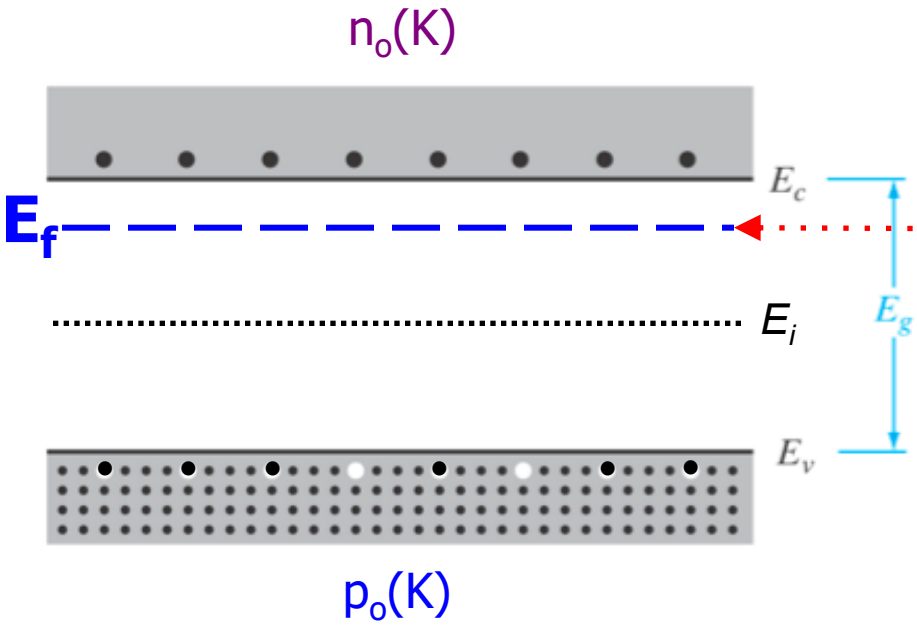


$$n = n_i e^{(F_n - E_i)/kT}$$

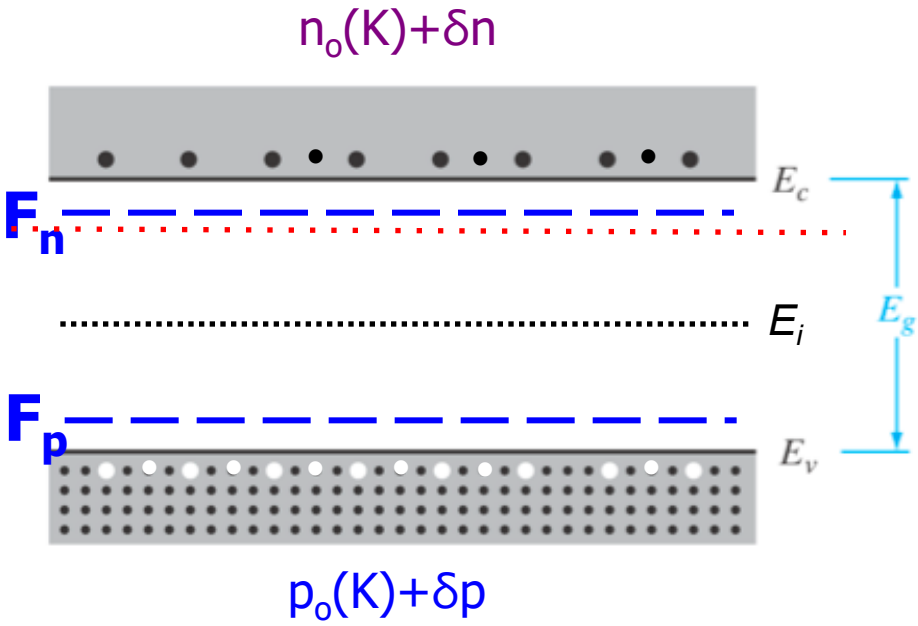
$$F_n - E_i = kT \ln\left(\frac{n}{n_i}\right)$$

Quasi-Fermi Levels in n-type Material

Equilibrium



With Excess Carriers



$$p = n_i e^{(E_i - F_p)/kT}$$

$$E_i - F_p = kT \ln\left(\frac{p}{n_i}\right)$$

Example: Steady-State Generation in Si

1) 10^{13} EHP/cm^3 optically created per microsecond

Therefore, $g_{opt} = \dfrac{10^{13}\,\text{EHP/cm}^3}{10^{-6}\,s} = 10^{19}$ EHP/cm^{3}s

2) $n_o = 10^{14}$ cm^{-3} in equilibrium (no illumination)

Therefore, $p_o = \dfrac{n_i^2}{n_o} = \dfrac{2.25 \times 10^{20}}{10^{14}}\text{cm}^{-3} = 2.25 \times 10^6$

3) The equilibrium Fermi level (no illumination) is calculated:

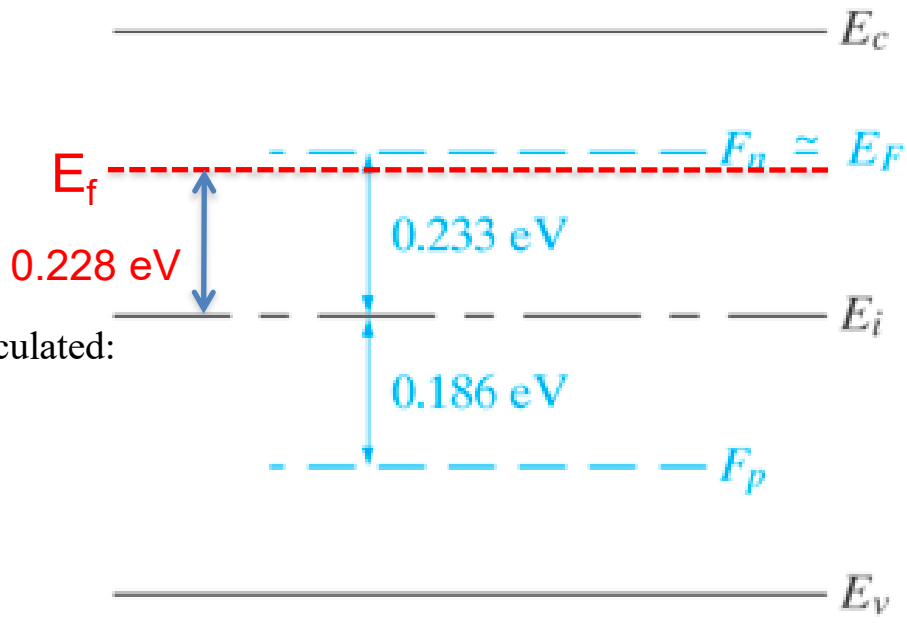
$n_o = 10^{14} = n_i e^{(E_f - E_i)/kT} = (1.5 \times 10^{10}) e^{(E_f - E_i)/kT}$

$E_f - E_i = 0.0259 \ln(6.7 \times 10^3) = 0.228 eV$

4) $\tau_n = \tau_p = 2\mu\text{sec}$

$g_{opt} = \dfrac{\delta n}{\tau}$

Therefore $\delta n = \delta p = g_{opt}\tau_n = 2 \times 10^{13}$



5) $n = n_o + \delta n = 1.2 \times 10^{14}\,\text{cm}^{-3} = (1.5 \times 10^{10}) e^{(F_n - E_i)/kT}$, kT=0.0259 eV

Therefore $F_n - E_i = 0.0259 \ln(8 \times 10^3) = 0.233 eV$

6) $p = p_o + \delta p = 2 \times 10^{13}\,\text{cm}^{-3} = (1.5 \times 10^{10}) e^{(E_i - F_p)/kT}$

Therefore $E_i - F_p = 0.0259 \ln(1.3 \times 10^3) = 0.186 eV$

Note: $np \neq n_i^2$ when excess carriers are present!

Fermi Level Comments

- The separation of the electron and hole quasi Fermi levels (F_n-F_p) provides a metric for the deviation from equilibrium
- The individual deviation F_n and F_p from E_F is a measure of how far the electron and hole populations are from equilibrium
- Depicting the quasi Fermi Levels on a band diagram gives a good representation of how a sample is locally deviating from equilibrium

$$np = \left(n_i e^{(F_n - E_i)/kT}\right)\left(n_i e^{(E_i - F_p)/kT}\right)$$

$$= n_i^2 e^{(F_n - E_i)/kT} e^{(E_i - F_p)/kT}$$

$$= n_i^2 e^{(F_n - E_i + E_i - F_p)/kT}$$

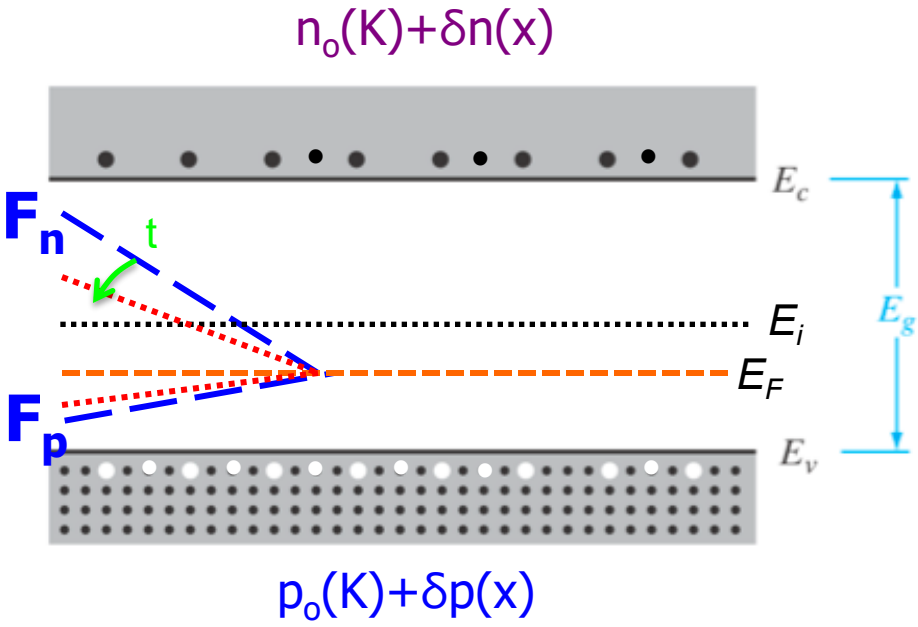
$$= n_i^2 e^{(F_n - F_p)/kT}$$



Valid for carrier densities such that quasi-Fermi level is 3kT away from band edge

Position and Time Dependence

- The quasi Fermi level can depend upon position (x)
- The quasi Fermi level will generally depend upon time (t) unless the device or material is under steady-state conditions



$$n = n_i e^{\left(F_n(x,t)-E_i\right)/kT}$$

$$p = n_i e^{\left(E_i-F_p(x,t)\right)/kT}$$

F_n, F_p more than 3kT away from E_c, E_v